

Malithi Mithsara

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 [Malithi Mithsara](#) |  [Malithi-gif](#)

Carbondale, Illinois - 62901, USA

OBJECTIVE

Seeking a challenging role in data science to utilize my expertise in computer science, statistics, and machine learning. Passionate about applying advanced analytical techniques to solve real-world problems and contribute to innovative projects at the intersection of data-driven insights and intelligent systems.

EXPERIENCE

- **Southern Illinois University.**  January 2024 - Current
Graduate Teaching Assistant Carbondale, USA
 - Teaching assistant for multiple computer science courses, including Operating Systems, Web Application Development, Machine Learning, and Soft Computing.
 - Coordinated ABET documentation by organizing, formatting, and maintaining course files (syllabi, assignments, assessments, and reports) to ensure compliance with accreditation requirements.
 - Researched large and small language models (LLMs/SLMs), utilizing fine-tuning and parameter-efficient techniques on multimodal data (text, vision, sensor) to address diverse real-world problems.
- **Informatics Institute of Technology.**  August 2023 - December 2023
Assistant Lecturer Colombo, Sri Lanka
 - Taught courses in object-oriented programming, fundamentals of computer systems, software development, and advanced mathematics for data science students in ICT and computer science.
 - Conducted and supervised research in computing and data science, guiding projects from proposal development to implementation, analysis, and final reporting. Supported research activities involving software development, data analysis, machine learning, and applied computing methods.
- **Rajarata University of Sri Lanka.**  August 2021 - August 2023
Temporary Lecturer Mihintale, Sri Lanka
 - Taught Web Development, Operating Systems, and Management Information Systems to second-year ICT and Computer Science students.
 - Conducted and supervised research-related activities in applied computing, information systems, software technologies, and computer systems. Guided research projects from proposal development to technical implementation, evaluation, and final reporting, with emphasis on practical problem-solving and real-world applications.
- **Open University of Sri Lanka.**  October 2021 - August 2023
Online Tutor Colombo, Sri Lanka
 - Taught courses in web development, computer security, Python programming, and mobile application development to undergraduate students.
- **Sri Lankan Institute of Information Technology, Sri Lanka.**  October 2020 - July 2021
Research Assistant Malabe, Sri Lanka
 - Supervised undergraduate research students at SLIIT while conducting independent research in computer vision, machine learning, and graph neural networks.
- **Department of Meteorology, Sri Lanka.**  March 2020 - October 2020
Information Technology Officer Colombo, Sri Lanka
 - Maintained and updated the Department of Meteorology's website.
- **University of Sri Jayewardenepura, Sri Lanka.**  January 2020 - March 2020
Research Assistant Colombo, Sri Lanka
 - Developed research objectives, prepared project proposals, conducted both individual and collaborative research projects, and supervised undergraduate research students.
- **University of Sri Jayewardenepura, Sri Lanka.**  June 2017 - December 2019
Teaching Assistant Colombo, Sri Lanka
 - Conducted regular tutorial and practical classes; supervised undergraduate students (1st–4th year) in Object-Oriented Programming (Java, C++, and C), Database Management Systems (DBMS), and Web Technologies.

EDUCATION

- **Southern Illinois University**
PhD in Computer Science
 - GPA: 4.00/4.00
 - Advisor: Prof. Ning Yan, Co-Advisor: Prof. A. R. Shahid
 - **University of Sri Jayewardenepura**
MSc in Computer Science
 - Grade: 3.30/4.00
 - **University of Sri Jayewardenepura**
Bachelor of Science General degree in Physical Science
 - GPA: 2.90/4.00

January 2024 - Present
Carbondale, USA

January 2018 - July 2020
Nugegoda, Sri Lanka

March 2014 - November 2016
Nugegoda, Sri Lanka

RESEARCH AND INTERESTS

My research focuses on human activity recognition (HAR) through large language models (LLMs) and small language models (SLMs), with a special emphasis on security and privacy challenges in LLM-integrated sensing systems. I explore how LLMs can be utilized to interpret complex sensor data for real-time activity analysis, while also investigating their vulnerabilities to various attacks, such as data poisoning, adversarial examples, and prompt injection attacks. My work aims to develop a privacy-preserving HAR pipeline that combines accurate sensing with secure and trustworthy AI-driven decision-making, especially in sensitive domains like healthcare and smart environments.

PROJECTS

- **Project A: [Automated Fake News Detection]**
Tools: [Jupyter notebook, Python]
 - Conducted comparative analysis of machine learning and deep learning models for fake news detection, including CNN, LSTM, DistilBERT, FineBERT, RoBERTa, ALBERT, ELECTRA, Gradient Boosting, and Random Forest.
 - **Project B: [AI-Powered YouTube Surveillance and Crime Detection]**
Tools: [Jupyter notebook, Python]
 - Developed Python-based code for detecting crime-related YouTube videos, utilizing the YouTube API to automatically download and preprocess video datasets for analysis and classification.

March 2025 - May 2025

March 2024

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] Malithi Wanniarachchi Kankanamge, et al. (2025). **Breaking Sensor-Text Alignment: Cross-Modal Attack on Contrastive Multimodal Human Activity Recognition**. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)*.
 - [C.1] Malithi Wanniarachchi Kankanamge, et al. (2025). **Development of a Fall Detection and Safety Communication System Using Small Language Models**. In *Proceedings of 42nd International Symposium on Automation and Robotics in Construction(ISARC)*, July 28-29, Montreal, Canada. DOI: 10.22260/ISARC2025/0077.
 - [C.1] WKM Mithsara, et al. (2024). **Zero-Shot Detection and Sanitization of Data Poisoning Attacks in Wearable AI Using Large Language Models**. In *the 23rd International Conference on Machine Learning and Applications(ICMLA)*, December 18-24, Miami, Florida, USA. DOI: 10.1109/ICMLA61862.2024.00233.
 - [C.1] Malithi Wanniarachchi Kankanamge, et al. (2024). **Intelligent Fall Detection and Emergency Response for Smart Homes Using Language Models**. In *23rd International Conference on Machine Learning and Applications (ICMLA)*, December 18-24, Miami, Florida, USA. DOI: 10.1109/ICMLA61862.2024.00037.
 - [C.1] WKM Mithsara, et al. (2024). **Large Language Model Integrated Healthcare Cyber-Physical Systems Architecture**. In *IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, July 02-04, Osaka, Japan. DOI: 10.1109/COMPSAC61105.2024.00228.
 - [C.1] Malithi Wanniarachchi Kankanamge, et al. (2024). **Poster: Leveraging Large Language Models for Zero-Shot Detection and Mitigation of Data Poisoning in Wearable AI Systems**. In *the NeurIPS 2024 Workshop GenAI4Health*, December 12, Vancouver, Canada.
 - [C.1] W.K.M Mithsara, et al. (2022). **Comparative Analysis of AI-Powered Approaches for Skeleton-Based Child and Adult Action Recognition in Multi-person Environment**. In *the International Conference on Computer Science and Software Engineering (CSASE)*, March 15 - 17.
 - [C.1] W. K. Malithi Mithsara, et al. (2020). **A Sociable Robotic Platform to Make Career Advices for Undergraduates**. In *HCI International 2020 - Late Breaking Papers: Multimodality and Intelligence* , July 19–24, Copenhagen, Denmark.
 - [J.1] W.K.M Mithsara, et al. (2020). **Blockchain-based Distributed Secure Crowdfunding and Decision-Making Platform for Large-scale Business Projects in Public and Private Sectors**. *European Modern Studies Journal*, Vol. 04, pp. 11.

SKILLS

- **Programming Languages:** JAVA, Python, C++, C, Language 5
- **Database Technologies:** MySQL, SQL (WAMP Server, XAMPP), Oracle
- **Web Technologies:** HTML, CSS, JavaScript, PHP, Web services (SOAP, REST), .NET
- **Mobile Development:** Android Application Development
- **Tools:** NetBeans, Eclipse, RapidMiner, Code Blocks, Maple, MATLAB, Minitab, Tableau, Unity
- **Research Skills:** Large Language Models and Small Language Models, Machine Learning and Deep Learning, Security and Privacy.

HONORS AND AWARDS

- **CVPR26 Broadening Participation Scholarship Award** June 2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 
 - Received travel support to attend CVPR 2026, present research, and participate in professional networking and research activities.
- **GPSC Travel and Conference Award** October 2024
Graduate and Professional Student Council 
 - Through funding for conference travel, presentations, and professional development, it enables students to showcase their work, network, and gain visibility in their field.
- **BSN Travel Award** October 2024
IEEE-EMBS International Conference on Body Sensor Networks 
 - Being selected in a competitive process backed by NSF and IEEE EMBS

LEADERSHIP EXPERIENCE

- **Secretary** February 2025 - February 2026
Sri Lankan Student Association (SLSA), SIUC 
 - Prepare, circulate, and maintain meeting agendas and minutes.
 - Coordinate international events for the society and the university.
 - Maintain documentation (e.g., constitutions, membership lists).
- **Committee Member** January 2016 - November 2016
Statistics Society of the University of Sri Jayewardenepura, Sri Lanka. 
 - Organizing events in collaboration with schools in Sri Lanka.
 - Assisted in planning and coordinating workshops, seminars, and competitions.
 - Supported fundraising activities and sponsorship coordination for society events.

PROFESSIONAL MEMBERSHIPS

- **Association for Computing Machinery:** Membership ID: 7351824 May 2024 - Present
- **IEEE:** Membership ID: 101772632 March 2026 - Present
- **Society of Women’s Engineering, SIUC** May 2025 - Present
- **Society of Women in Cybersecurity** Membership ID: 92375949 March 2026 - Present

ADDITIONAL INFORMATION

Languages: English (Intermediate level), **Interests:** Swimming, Badminton

REFERENCES

1. **Prof. Ning Yang,**
Professor in Information Technology, Faculty of Information Technology,
Southern Illinois University, Carbondale, USA.
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Relationship: [Advisor]
2. **Prof. A. R. Shahid**
Department of Computer Science,
School of Computing, Southern Illinois University, Carbondale, USA.
Email: abdurrahmanbin.shahid@siu.edu
Phone: +1-786-397-2558
Relationship: [Co-Advisor]
3. **Prof. R.G.N. Meegama,**
Senior Professor in Computer Science, Faculty of Applied Sciences,
University of Sri Jayewardenepura, Sri Lanka.
Email: rgn@sci.sjp.ac.lk
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Relationship: [Teacher]